Two-player handheld: reference design (REF-A)

RP2350B, 3.5in 320x480 IPS (ST7796, SPI), 16 membrane contacts under DS Lite silicone,
MAX98357A audio, one Li-ion cell on a BQ24074 power path, TPS63001 3V3 buck-boost.

Every sheet is generated from board/design.py by tools/build_sch.py. Edit design.py, not the sheets.

USB_Charger

usb_charger.kicad_sch

USB-C sink, ESD, BQ24074 power path
charger, battery connector, NTC

Power

power.kicad_sch

TPS63001 3V3 buck-boost, power button
latch, battery and button sensing

MCU

mcu.kicad_sch

RP2350B, 12 MHz crystal,
16 MB QSPI flash, USB series resistors

MCU_Power

mcu_power.kicad_sch

RP2350B decoupling and its
internal 1.1 V switcher

Debug

debug.kicad_sch

SWD and UART connectors, BOOTSEL
and RESET buttons, error LED

Display

display.kicad_sch

48-pin solder FPC of the 3.5in IPS panel,
AP3156 backlight driver

Audio

audio.kicad_sch

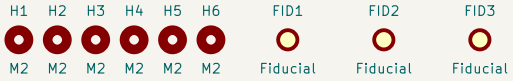
MAX98357A I2S class D amplifier,
speaker pads

Buttons

buttons.kicad_sch

16 membrane contacts,
one GPIO each

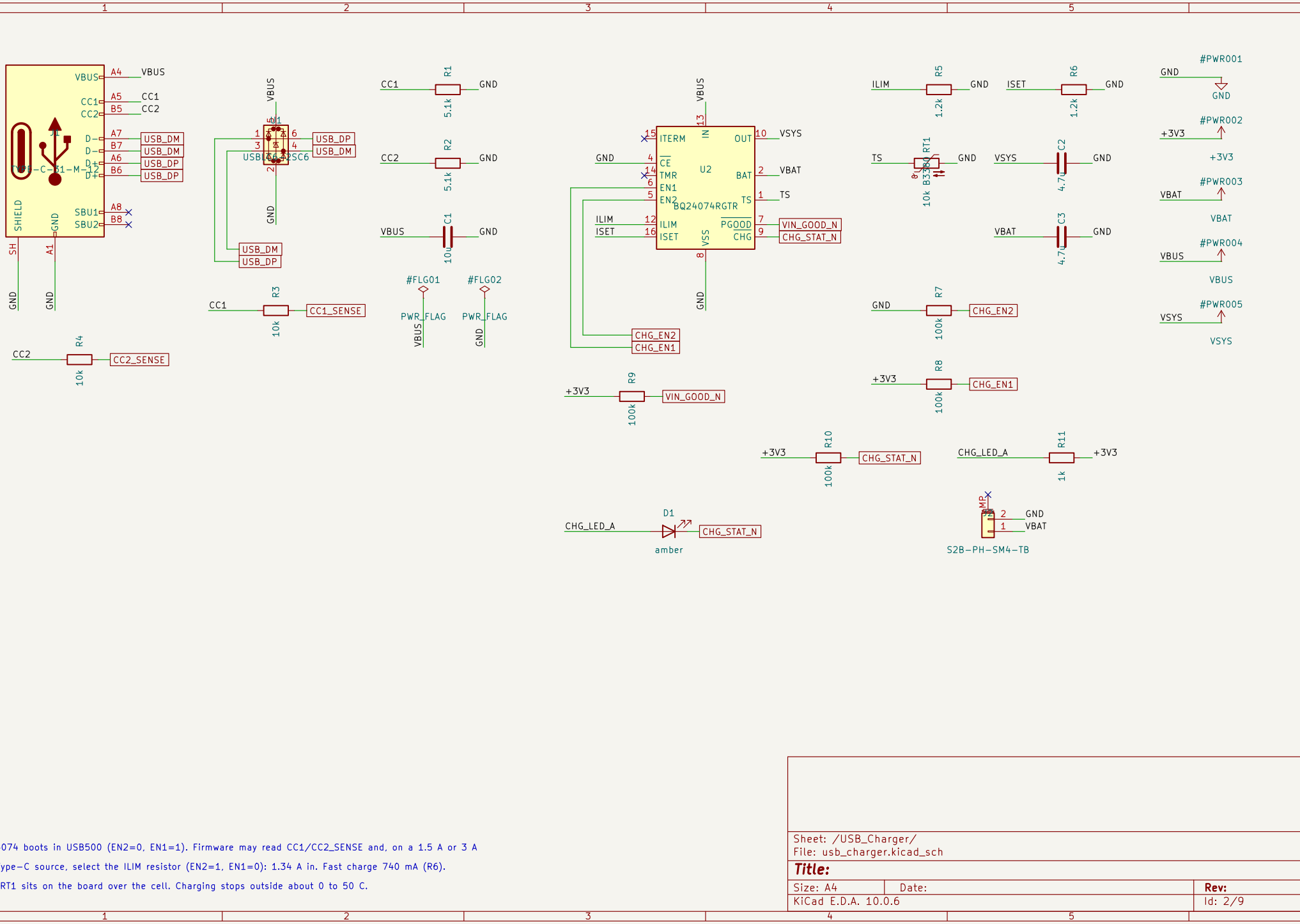
Mechanical: M2 mounting holes at positions set by the case; fiducials for assembly.



Sheet: /
File: battler.kicad_sch

Title:

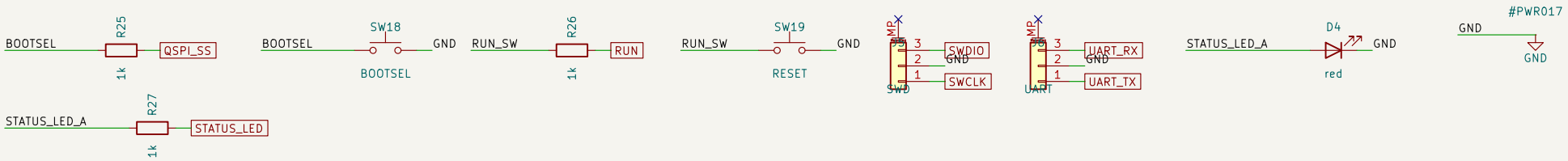
Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 1/9



BQ24074 boots in USB500 (EN2=0, EN1=1). Firmware may read CC1/CC2_SENSE and, on a 1.5 A or 3 A Type-C source, select the ILIM resistor (EN2=1, EN1=0): 1.34 A in. Fast charge 740 mA (R6).

RT1 sits on the board over the cell. Charging stops outside about 0 to 50 C.

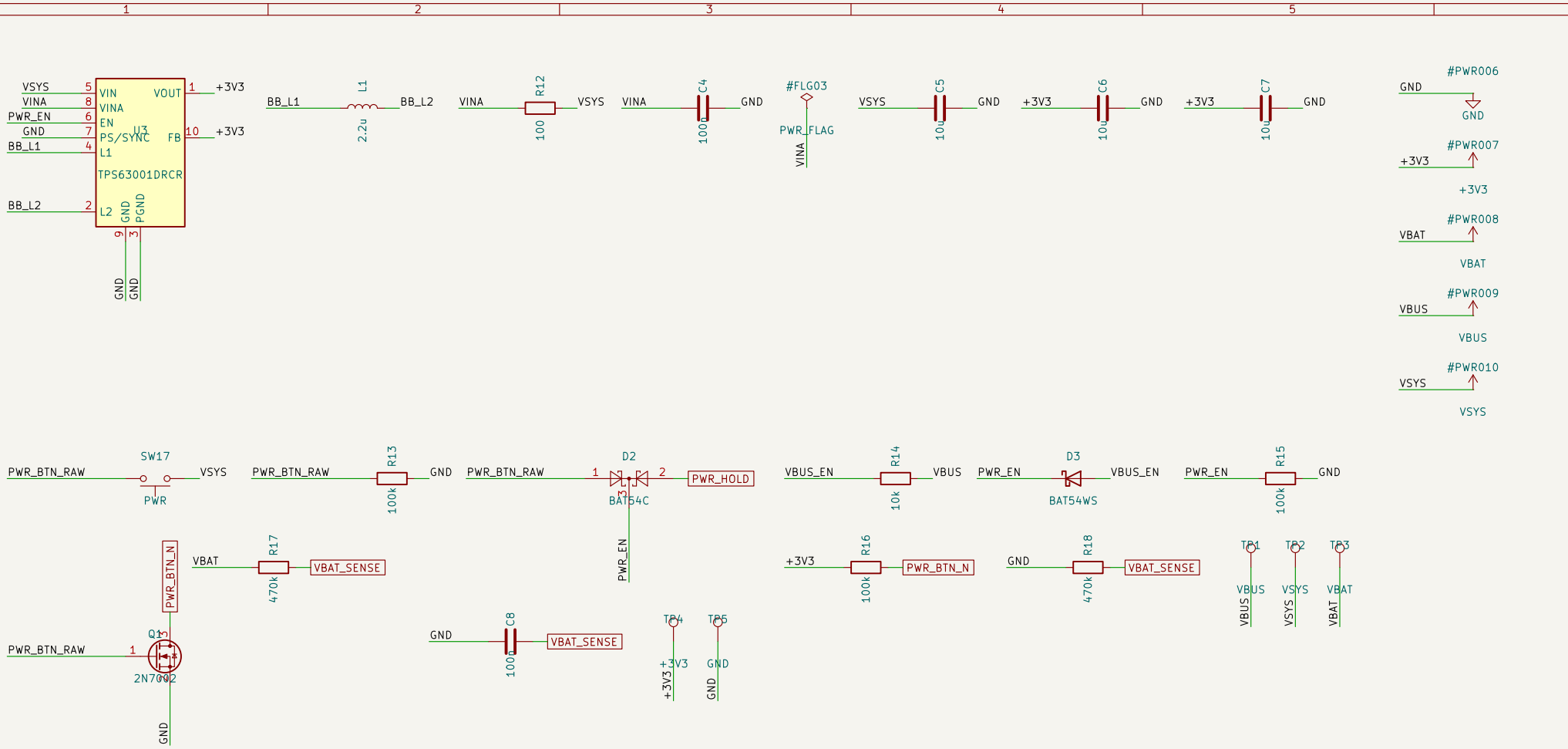
Sheet: /USB_Charger/		
File: usb_charger.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/9



J5 SWD and J6 UART follow the Raspberry Pi 3-pin debug connector pinout (Debug Probe cable).

SW18 BOOTSEL and SW19 RESET sit under pinholes in the lid. D4 is the error LED.

Sheet: /Debug/		
File: debug.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/9



Power: the button, the MCU (PWR_HOLD) or USB each raise PWR_EN. Releasing the button with PWR_HOLD low turns the device off. A reset drops PWR_HOLD, so a crash turns it off.

Q1 lets the MCU see the button (PWR_BTN_N low = pressed) without exposing a pin to VSYS.

Sheet: /Power/
File: power.kicad_sch

Title:

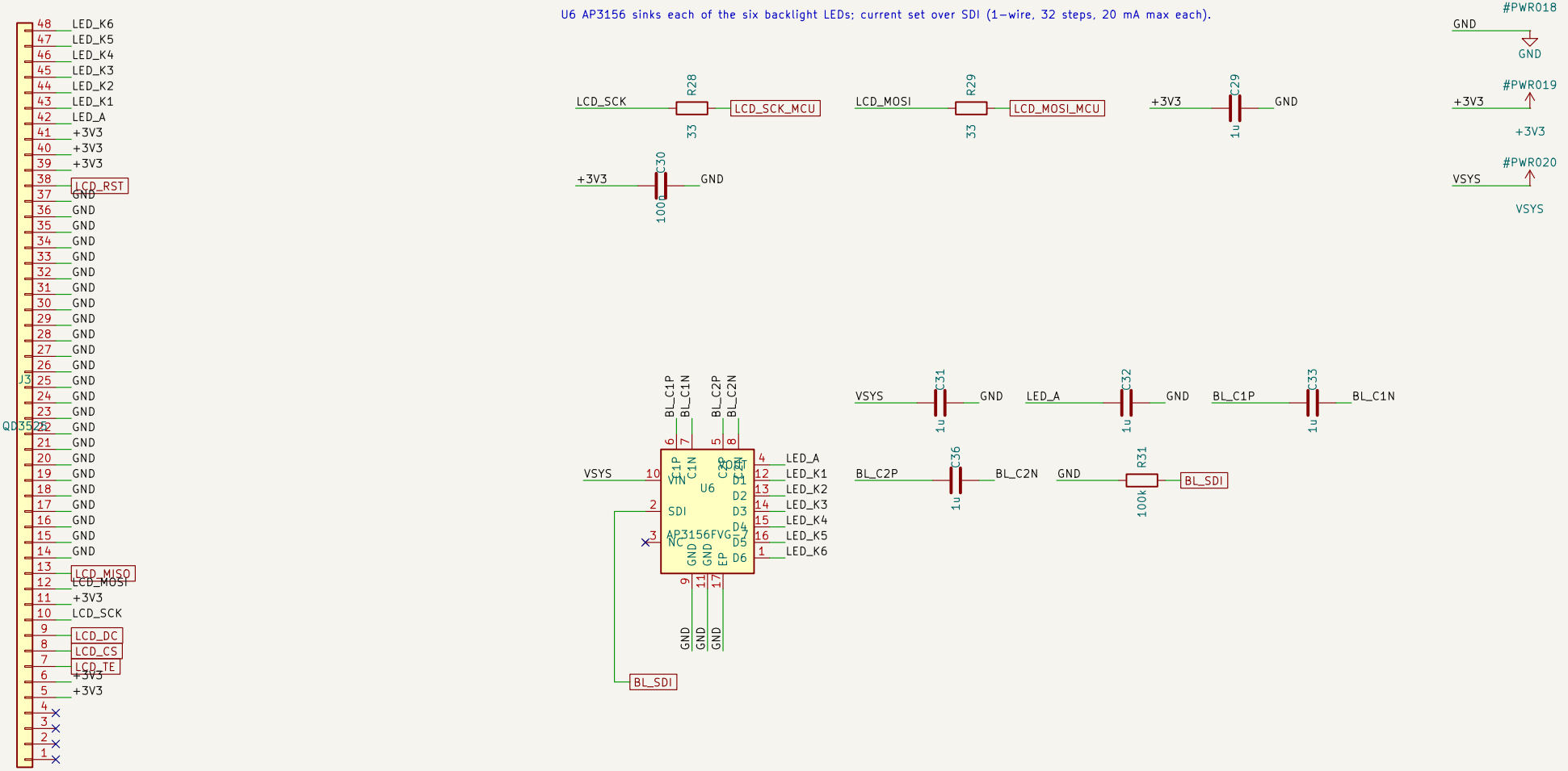
Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 4/9

QD3525 3.5in IPS panel (ST7796U, 4-line SPI). Its 48-pin FPC folds under the panel and is soldered to J3.

Unused bus pins are tied as the panel maker's own module ties them: DB0-17 and RGB low, RD and IM2-0 high.

U6 AP3156 sinks each of the six backlight LEDs; current set over SDI (1-wire, 32 steps, 20 mA max each).



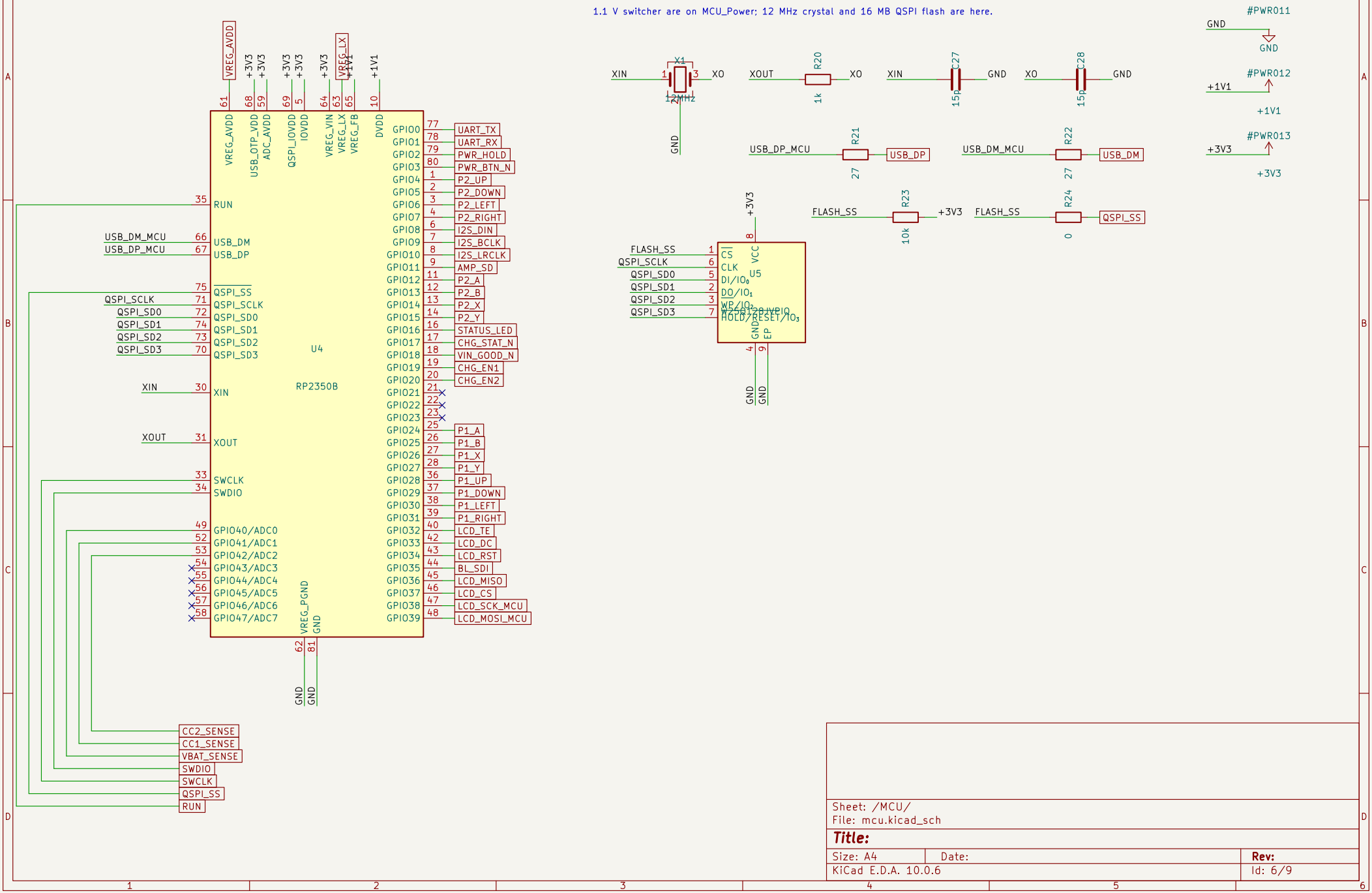
Sheet: /Display/
File: display.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/9

Core from Raspberry Pi's RP2350B minimal design (MIT). Supply decoupling and the

1.1 V switcher are on MCU_Power; 12 MHz crystal and 16 MB QSPI flash are here.

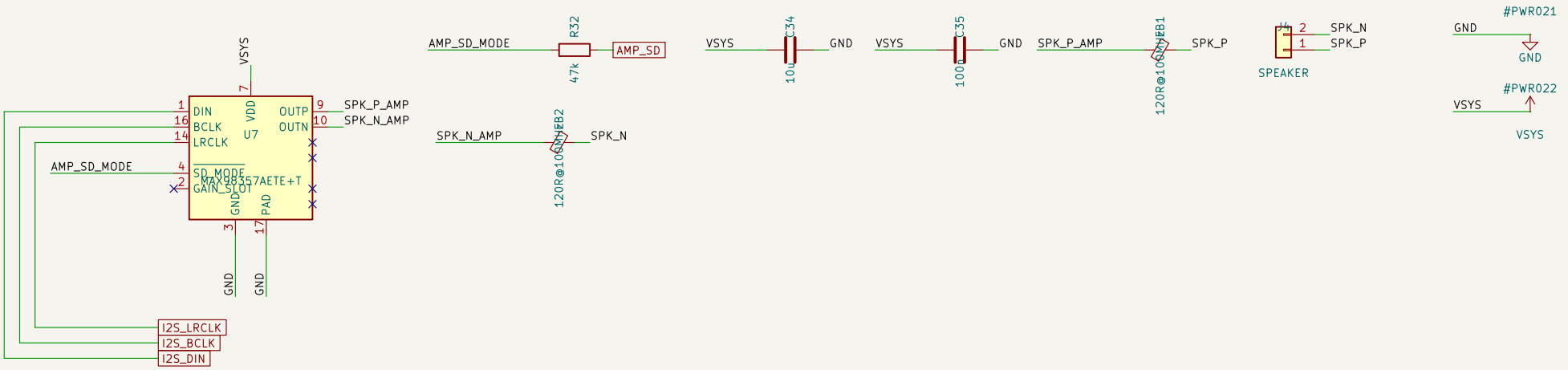


Sheet: /MCU/
File: mcu.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

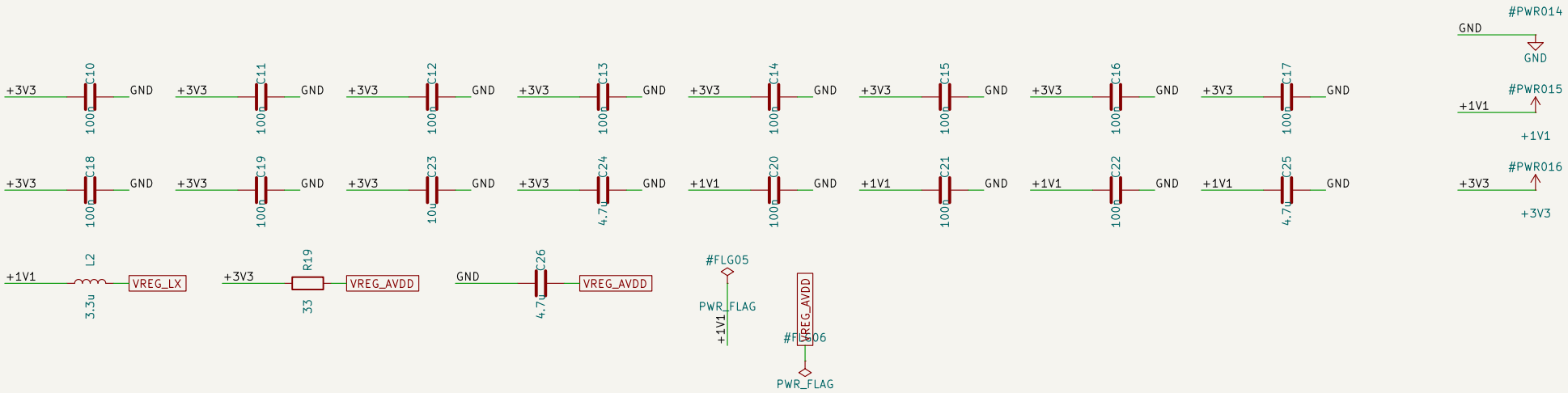
KiCad E.D.A. 10.0.6 | Id: 6/9



SD_MODE through R32 sits near 2.2 V: left channel. Firmware writes the mono mix to both slots.

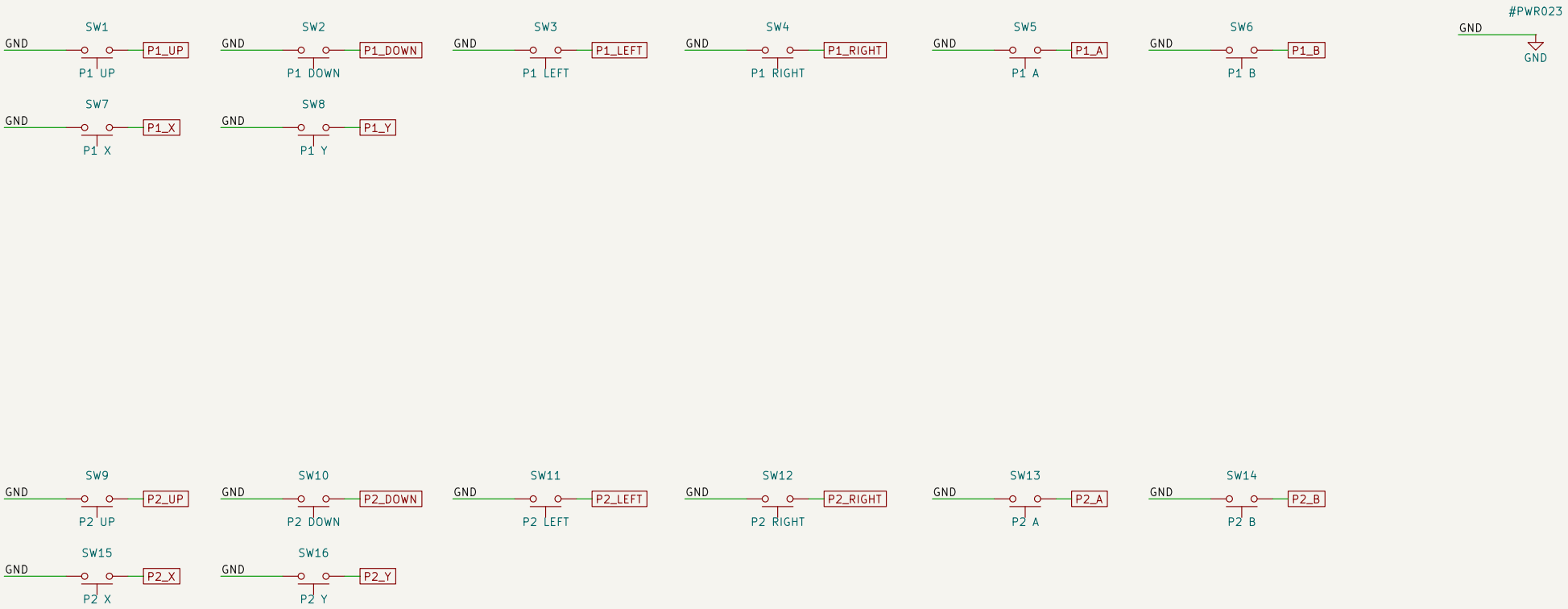
GAIN_SLOT open: 9 dB. 8 ohm 1 W speaker, its leads soldered to the J4 pads from below.

Sheet: /Audio/	
File: audio.kicad_sch	
Title:	
Size: A4	Date:
KiCad E.D.A. 10.0.6	Id: 7/9



RP2350B supply, as in Raspberry Pi's minimal design: 100 nF at every IOVDD, DVDD, ADC, OTP and QSPI pin, 4.7 uF at the regulator's input and output, and L2 (a polarity-marked 3.3 uH inductor, fit as drawn).

Sheet: /MCU_Power/	
File: mcu_power.kicad_sch	
Title:	
Size: A4	Date:
KiCad E.D.A. 10.0.6	Rev: Id: 8/9



Bare ENIG contacts (DSL_button_4ring) under DS Lite silicone; the firmware enables pull-ups.

Sheet: /Buttons/
File: buttons.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 9/9